AI-Powered Student Study Planner

Project Synopsis

1. Introduction

AI-Powered Student Study Planner is a proposed Flutter-based mobile application designed to organize preparation across multiple subjects and academic deadlines. The system creates personalized study schedules based on subjects, upcoming exams, and available daily study time to make academic planning dynamic, efficient, and structured.

2. Problem Statement

Students struggle to organize study routines effectively across multiple subjects with tight deadlines, often leading to poor time management, last-minute cramming, and burnout. Manual planning using paper schedules or basic calendars lacks dynamic adaptability when tasks are delayed or exam priorities change. Students require an automated, digital solution that calculates and adjusts study schedules dynamically.

3. Objectives

- Develop a mobile application for automated study schedule generation.
- Provide an interactive dashboard to monitor overall study progress and upcoming deadlines.
- Enable subject and task management with difficulty weightages and estimated preparation hours.
- Utilize AI algorithms to convert user availability, task deadlines, and subject difficulty into optimized study routines.
- Provide interactive calendar views to track and toggle completion of daily study blocks.
- Deliver a responsive, user-friendly interface with structured error handling.

4. Proposed System

The application provides an interface through which users authenticate and access academic planning features. Users input subjects, assign difficulty levels, and log assignments or exams with target deadlines. Based on available daily study hours, the AI generation engine calculates optimized, color-coded study schedules. Users can view these blocks in an interactive calendar and mark progress, triggering real-time routine updates.

5. Major Modules

Module	Description
Authentication	User sign-up, login, and initial onboarding (setting daily available study hours).

Dashboard	Central overview displaying daily study target progress, quick navigation, and upcoming exam deadlines.
Subject Manager	Interface to add, update, or remove subjects with difficulty weightage ratings (1–5).
Task & Exam Manager	Interface to record assignments, topics, deadlines, and estimated preparation time.
AI Schedule Generator	Engine that processes tasks, deadlines, and time constraints to generate dynamic study routines.
Interactive Calendar	Calendar interface displaying scheduled study blocks with status toggling (completed, pending).
Profile & Settings	Manage user preferences, adjust daily available hours, and reset schedules.

6. User Interface / Screens

The application contains six main screens:

- **Login / Register Screen:** Handles user authentication and daily study hour configuration.
- **Dashboard Screen:** Highlights high-level study metrics, upcoming deadlines, and primary actions.
- **Subject Manager Screen:** Provides CRUD operations for academic courses and difficulty ratings.
- **Task Input Screen:** Logs assignments, exams, topic breakdowns, and deadline targets.
- **AI Generator Screen:** Prompts the generation engine to compute optimized time-block allocations.
- **Interactive Calendar Screen:** Displays daily/weekly scheduled study blocks using a calendar UI.

7. Technology Stack

- **Frontend / UI:** Flutter and Dart (Material Design 3)
- **State Management:** Flutter Riverpod
- **Navigation:** Go Router
- **Backend & Storage:** Firebase Authentication, Cloud Firestore (or local SQLite/Hive storage)
- **AI Integration:** Google Generative AI (Gemini API)
- **Version Control:** Git and GitHub

8. Functional Requirements

- The system shall authenticate authorized users securely.
- The system shall allow users to input courses, difficulty weightages, and deadline constraints.
- The system shall accept daily user availability metrics and compute optimized study schedules via AI integration.

- The system shall store, retrieve, and update study schedules and task completion states in real-time.
- The application shall validate user forms and handle API or database errors gracefully.
- The application shall provide smooth navigation across all six core screens.

9. Non-Functional Requirements

The application shall maintain a responsive, intuitive, and consistent UI design across mobile screen sizes. It shall handle operational errors gracefully without crashing, write organized modular code, and follow version control best practices on GitHub.

10. Methodology

The project follows an iterative development methodology:

1. Finalize application requirements, database schema, and UI screen workflows.
2. Design screen mockups and project directory layout.
3. Build Flutter UI modules, state providers, and navigation routing.
4. Integrate Firebase backend services and Gemini API scheduling endpoints.
5. Perform unit, integration, and UI testing across application features.
6. Commit code routinely to GitHub for collaborative project tracking.

11. Expected Outcomes

- A fully functional Flutter application for student academic planning.
- Automated generation of personalized study schedules based on student inputs.
- Structured tracking of subject deadlines, task difficulty, and study progress.
- Reduced planning effort and elimination of last-minute exam prep stress.

12. Future Scope

Future updates can expand the application with push notifications for study block start times, social study group integration, automated performance analytics, Pomodoro timer integration, and institution-level syllabus auto-import.

13. Conclusion

AI-Powered Student Study Planner provides a modern, automated solution for academic time management. By combining Flutter cross-platform mobile development with AI schedule optimization, the system simplifies routine creation, prioritizes high-impact subjects, and helps students achieve academic goals.